STT 861: Theory of Probability & Statistics I

Chapter 3.4–3.6: Exponential Families, Location–Scale Families, Inequalities and Identities

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Reference: Casella & Berger Ch. 3, Sections 3.4, 3.5 and 3.6

Where we are: from individual families to structure

We now have a catalogue (§3.1–3.3): Binomial, Poisson, Gamma, Normal, Beta, Cauchy, ... Each was studied one at a time.

Today: what do families have in common?

- **§3.4 Exponential families:** one form covers many families; moments come from a single derivative.
- **§3.5 Location–scale families:** how shifting and rescaling generate a family.
- **§3.6 Inequalities & identities:** bounds on probabilities and tricks for moments.

Why this matters later

- Sufficient statistics, MLEs and generalized linear models (Chapters 6–8, STT 862) are cleanest for exponential families.
- Location–scale structure explains pivotal quantities and why $\bar{X}$, s behave nicely.
- Inequalities are the workhorse of convergence proofs (Ch. 5).

§3.4 Motivation: many families, one pattern

Compare three densities:

$$\text{Poisson: } \frac{1}{x!}\, e^{-\lambda}\, \exp\{x \log \lambda\}$$

$$\text{Exponential: } \frac{1}{\beta}\, \exp\{-x/\beta\}$$

$$\text{Bernoulli: } (1-p)\exp\{x \log \tfrac{p}{1-p}\}$$

Each is (function of x) $\times$ (function of θ) $\times$ $\exp\{(\text{function of } \theta) \cdot (\text{function of } x)\}$.

Payoff of recognizing the pattern

Once a family fits this form we get, for free:

- $\mathbb{E}$ and Var by differentiating $\log c(\theta)$, with no integrals or sums;
- the mgf of the sufficient statistic;
- good behaviour of estimators and tests later in the course.

Think (1 min)

Is the Uniform$(0, \theta)$ density $\frac{1}{\theta} I(0 < x < \theta)$ of this form? *(We will see: no.)*

Definition: exponential family

Definition (CB 3.4.1)

A family of pdfs or pmfs is an **exponential family** if it can be written

$$f(x \mid \boldsymbol{\theta}) = h(x)\, c(\boldsymbol{\theta}) \exp\Big\{ \sum_{i=1}^{k} w_i(\boldsymbol{\theta})\, t_i(x) \Big\},$$

where $h(x) \geq 0$ and $t_1, \ldots, t_k$ depend on x only, and $c(\boldsymbol{\theta}) \geq 0$ and $w_1, \ldots, w_k$ depend on $\boldsymbol{\theta}$ only.

- **Key requirement:** the x-part and the θ-part *separate*; in particular the **support cannot depend on** θ (put any indicator into $h(x)$).
- $c(\theta)$ is a normalizing constant: $c(\theta) = \left[\int h(x) e^{\sum w_i t_i(x)} dx \right]^{-1}$.
- The same family can be written in several equivalent ways.

Checklist

(1) support free of θ; (2) take $\log f$ and see whether the x–θ cross terms are a sum of products $w_i(\theta) t_i(x)$.

Examples: Binomial and Poisson

Binomial(n, p), n known

$$f(x \mid p) = \binom{n}{x} p^x (1-p)^{n-x} = \underbrace{\binom{n}{x}}_{h(x)} \underbrace{(1-p)^n}_{c(p)} \exp\Big\{ \underbrace{\log \tfrac{p}{1-p}}_{w(p)} \underbrace{x}_{t(x)} \Big\}.$$

The support $\{0, \ldots, n\}$ does not depend on p.

Poisson(λ)

$$f(x \mid \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} = \underbrace{\tfrac{1}{x!}}_{h(x)} \underbrace{e^{-\lambda}}_{c(\lambda)} \exp\Big\{ \underbrace{\log \lambda}_{w(\lambda)} \underbrace{x}_{t(x)} \Big\}.$$

Pattern

Binomial, Poisson, Geometric, Negative Binomial (fixed r) and Bernoulli all have $t(x) = x$ with w a function of the parameter.

Examples: Normal and Gamma (two-parameter families)

Normal(μ, σ^2)

$$f = \underbrace{1}_{h} \underbrace{\frac{e^{-\mu^2/2\sigma^2}}{\sqrt{2\pi}\sigma}}_{c(\mu,\sigma^2)} \exp\left\{ \tfrac{1}{\sigma^2}\left(-\tfrac{x^2}{2}\right) + \tfrac{\mu}{\sigma^2}\, x \right\}$$

$k = 2$: $t_1 = -x^2/2$, $w_1 = 1/\sigma^2$; $t_2 = x$, $w_2 = \mu/\sigma^2$.

Gamma(α, β)

$$f = \underbrace{I(x > 0)}_{h} \underbrace{\frac{1}{\Gamma(\alpha)\beta^\alpha}}_{c(\alpha,\beta)} \exp\left\{ (\alpha - 1)\log x - \tfrac{1}{\beta}\, x \right\}$$

$t_1 = \log x$, $w_1 = \alpha - 1$; $t_2 = x$, $w_2 = -1/\beta$.

Also exponential families

Exponential, χ^2, Beta (with $t = \log x$, $\log(1 - x)$), Lognormal, Double exponential with μ known. Sufficient statistics will turn out to be $\sum t_i(X_j)$.

Non-examples

Uniform$(0, \theta)$

$f(x \mid \theta) = \frac{1}{\theta} I(0 < x < \theta)$.

- The indicator $I(x < \theta)$ couples x and θ: it cannot be written as $h(x) \times c(\theta) \times \exp\{\cdots\}$.
- The support depends on θ.

$\Rightarrow$ **not** an exponential family.

Cauchy(θ)

$f(x \mid \theta) = \frac{1}{\pi[1+(x-\theta)^2]}$. Here $\log f = -\log \pi - \log(1 + (x-\theta)^2)$ does not split into sums of products $w(\theta)t(x)$.
$\Rightarrow$ **not** an exponential family.

A subtle case: curved exponential families

$N(\mu, \mu^2)$ (variance tied to mean): $w_1 = \mu^{-2}$ and $w_2 = \mu^{-1}$ are functions of the single parameter μ, so $d = 1 < k = 2$. It is exponential-type but *curved*; many nice properties (e.g. completeness, §6) fail. For *full* families the parameter dimension d equals k.

Theorem: moments of $t(X)$ from $\log c(\theta)$

Theorem (CB 3.4.2, one-parameter, $k = 1$)

If $f(x \mid \theta) = h(x)c(\theta)\exp\{w(\theta)t(x)\}$, then

$$\mathbb{E}\big[w'(\theta)\, t(X)\big] = -\frac{d}{d\theta}\log c(\theta),$$

$$\mathrm{Var}\big[w'(\theta)\, t(X)\big] = -\frac{d^2}{d\theta^2}\log c(\theta) - \mathbb{E}\big[w''(\theta)\, t(X)\big].$$

Idea of the proof

Differentiate $\int h(x)c(\theta)e^{w(\theta)t(x)}dx = 1$ once in θ: $\frac{c'}{c} + w'\mathbb{E}t(X) = 0$; differentiate again for the variance.

Natural parameterization

With $\eta = w(\theta)$: $f(x \mid \eta) = h(x)c^*(\eta)e^{\eta t(x)}$ and then simply

$$\mathbb{E}t(X) = -\frac{d}{d\eta}\log c^*(\eta), \quad \mathrm{Var}\, t(X) = -\frac{d^2}{d\eta^2}\log c^*(\eta), \quad \mathbb{E}e^{st(X)} = \frac{c^*(\eta)}{c^*(\eta + s)}.$$

Examples: moments with no integration

Poisson(λ)

$c(\lambda) = e^{-\lambda}$, $w = \log\lambda$, $t = x$. $w' = \frac{1}{\lambda}$, $w'' = -\frac{1}{\lambda^2}$.

- $\mathbb{E}\left[\frac{X}{\lambda}\right] = -\frac{d}{d\lambda}(-\lambda) = 1 \Rightarrow \mathbb{E}X = \lambda$.
- $\mathrm{Var}\left[\frac{X}{\lambda}\right] = 0 - \mathbb{E}\left[-\frac{X}{\lambda^2}\right] = \frac{1}{\lambda} \Rightarrow \mathrm{Var}\,X = \lambda$.

Gamma(α, β), α known

$c(\beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha}$, $w = -\frac{1}{\beta}$, $t = x$; $w' = \frac{1}{\beta^2}$, $w'' = -\frac{2}{\beta^3}$; $\log c = -\alpha\log\beta + \text{const}$.

- $\mathbb{E}\left[\frac{X}{\beta^2}\right] = \frac{\alpha}{\beta} \Rightarrow \mathbb{E}X = \alpha\beta$.
- $\mathrm{Var}\left[\frac{X}{\beta^2}\right] = -\frac{\alpha}{\beta^2} + \frac{2\alpha}{\beta^2} = \frac{\alpha}{\beta^2} \Rightarrow \mathrm{Var}\,X = \alpha\beta^2$.

Bernoulli(p) in natural form

$\eta = \log\frac{p}{1-p}$, $c^*(\eta) = \frac{1}{1+e^\eta}$: $\mathbb{E}X = -\frac{d}{d\eta}\left[-\log(1+e^\eta)\right] = \frac{e^\eta}{1+e^\eta} = p$, $\mathrm{Var}\,X = \frac{e^\eta}{(1+e^\eta)^2} = p(1-p)$. (The logistic function appears: basis of logistic regression.)

Summary: common exponential families

Family	$t(x)$	**Natural parameter(s)**	$\mathbb{E}t(X)$	**Note**
Bernoulli(p)	x	$\log \frac{p}{1-p}$	p	logistic link
Binomial(n, p)	x	$\log \frac{p}{1-p}$	np	n known
Poisson(λ)	x	$\log \lambda$	λ	log link
Exponential(β)	x	$-1/\beta$	β	
Gamma(α, β)	$(\log x,\ x)$	$(\alpha - 1,\ -1/\beta)$	$(\psi(\alpha) + \log \beta,\ \alpha\beta)$	2 parameters
Normal(μ, σ^2)	$(x,\ x^2)$	$(\frac{\mu}{\sigma^2},\ -\frac{1}{2\sigma^2})$	$(\mu,\ \mu^2 + \sigma^2)$	2 parameters
Beta(α, β)	$(\log x,\ \log(1-x))$	$(\alpha,\ \beta)$		2 parameters

Not exponential families

Uniform(a, b) and any family whose support depends on the parameter; Cauchy; Weibull with unknown shape γ; Hypergeometric in N.

Activity 1 — Exponential families

Problems

(a) Write Exponential(β), $f(x) = \frac{1}{\beta} e^{-x/\beta}$, in natural form $h(x)c^*(\eta)e^{\eta t(x)}$. Identify η and $c^*(\eta)$, and use $-\frac{d}{d\eta} \log c^*$ and $-\frac{d^2}{d\eta^2} \log c^*$ to find $\mathbb{E}X$ and $\operatorname{Var} X$.

(b) Poisson(λ): find η and $c^*(\eta)$, and verify $\mathbb{E}X = \operatorname{Var} X = \lambda$ in the same way.

(c) Explain why Uniform$(0, \theta)$ is not an exponential family.

(d) For the Binomial(n, p) with n known, use the theorem to show $\mathbb{E}X = np$.

Activity 1 — Solutions

(a) $t(x) = x$, $\eta = -\frac{1}{\beta} < 0$, $c^*(\eta) = -\eta$, $h = I(x > 0)$. $\log c^* = \log(-\eta)$: $\mathbb{E}X = -\frac{d}{d\eta}\log(-\eta) = -\frac{1}{\eta} = \beta$; $\mathrm{Var}\, X = -\frac{d^2}{d\eta^2}\log(-\eta) = \frac{1}{\eta^2} = \beta^2$.

(b) $\eta = \log\lambda$, $c^*(\eta) = \exp\{-e^\eta\}$, $\log c^* = -e^\eta$. $\mathbb{E}X = e^\eta = \lambda$, $\mathrm{Var}\, X = e^\eta = \lambda$.

(c) The support $(0, \theta)$ depends on θ: $I(0 < x < \theta)$ cannot be factored into a function of x times a function of θ.

(d) $c(p) = (1-p)^n$, $w = \log\frac{p}{1-p}$, $w' = \frac{1}{p(1-p)}$, $-\frac{d}{dp}\log c = \frac{n}{1-p}$. So $\mathbb{E}\left[\frac{X}{p(1-p)}\right] = \frac{n}{1-p}$ and $\mathbb{E}X = np$.

§3.5 Motivation: change of units

A temperature example

Daily high temperature Z (in standardized units) has density $f(z)$.

- Shift the origin: $X = Z + \mu$ (a different reference point).
- Change the unit: $X = \sigma Z$ (Celsius $\to$ Fahrenheit multiplies by $\frac{9}{5}$).
- Both: $X = \sigma Z + \mu$.

The *shape* of the distribution is unchanged; only its position and spread change.

The idea

From *one* pdf f we can generate a whole family indexed by μ (location) and $\sigma > 0$ (scale). Probabilities for any member reduce to probabilities for the standard member f.

Think

Which families we have seen look like "one standard shape, shifted and stretched"? *(Normal, Cauchy, double exponential, exponential, uniform.)*

Theorem: shifting and rescaling a pdf gives a pdf

Theorem (CB 3.5.1)

Let $f(x)$ be any pdf, μ any real number and $\sigma > 0$. Then $g(x \mid \mu, \sigma) = \frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right)$ is also a pdf.

Proof

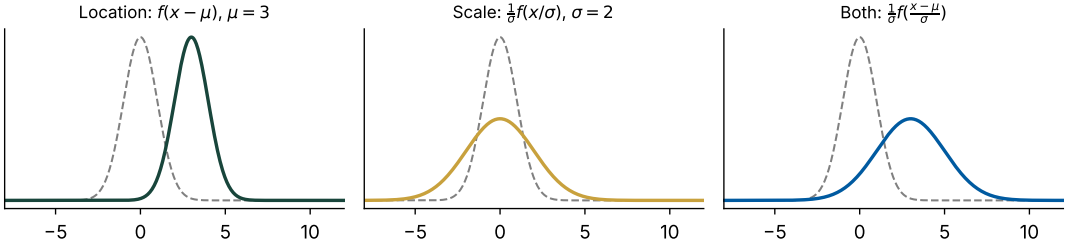
$g \geq 0$, and with $u = (x-\mu)/\sigma$, $dx = \sigma\, du$:
$\int \frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right) dx = \int f(u)\, du = 1$. ■

Definitions (CB 3.5.2–3.5.5)

Given a "standard" pdf f:

- **Location family:** $\{f(x-\mu) : \ \mu \in \mathbb{R}\}$.
- **Scale family:** $\{\frac{1}{\sigma} f(x/\sigma) : \ \sigma > 0\}$.
- **Location–scale family:** $\{\frac{1}{\sigma} f(\frac{x-\mu}{\sigma})\}$.

Illustration: location, scale and both



Reading the picture

The dashed curve is the standard pdf f. A location parameter slides it left/right; a scale parameter stretches it horizontally (and compresses it vertically so the area stays 1).

Examples

- $f =$ standard normal: $N(\mu, \sigma^2)$.
- $f = \frac{1}{\pi} \frac{1}{1+z^2}$: Cauchy$(\mu, \sigma)$.
- $f = \frac{1}{2} e^{-|z|}$: double exponential(μ, σ).

Examples of location, scale and location–scale families

Family	**Type**	**Standard pdf f**
Normal$(\mu, 1)$	location	$\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$
Normal$(0, \sigma^2)$	scale	$\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$
Normal(μ, σ^2)	location–scale	$\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$
Exponential(β)	scale $(\sigma = \beta)$	e^{-z}, $z > 0$
Uniform$(0, \theta)$	scale $(\sigma = \theta)$	1 on $(0, 1)$
Uniform(a, b)	location–scale $(\mu = a, \sigma = b - a)$	1 on $(0, 1)$
Cauchy(μ, σ)	location–scale	$\frac{1}{\pi(1+z^2)}$
Double exponential(μ, σ)	location–scale	$\frac{1}{2} e^{-\lvert z \rvert}$
Gamma(α, β), α fixed	scale $(\sigma = \beta)$	Gamma$(\alpha, 1)$ pdf

Caution

Uniform$(0, \theta)$ is a scale family but not an exponential family; Normal is both; Gamma(α, β) with *both* parameters free is exponential but not location–scale.

Representation theorem: $X = \sigma Z + \mu$

Theorem (CB 3.5.6)

Let f be any pdf, $\mu \in \mathbb{R}$, $\sigma > 0$. Then X has pdf $\frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right)$ **if and only if** there is a random variable Z with pdf f such that $X = \sigma Z + \mu$.

Proof ($\Leftarrow$)

$F_X(x) = P(\sigma Z + \mu \le x) = F_Z\left(\frac{x-\mu}{\sigma}\right)$; differentiate: $f_X(x) = \frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right)$. ($\Rightarrow$): reverse the steps with $Z = (X - \mu)/\sigma$.

Consequences

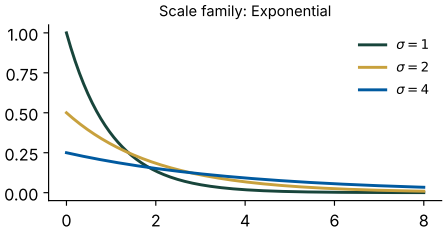
- $P(X \le x) = P\left(Z \le \frac{x-\mu}{\sigma}\right)$
- $\mathbb{E}X = \sigma \mathbb{E}Z + \mu$, $\operatorname{Var} X = \sigma^2 \operatorname{Var} Z$ (when they exist)
- $(X - \mu)/\sigma$ has a distribution **free of the parameters**: a *pivot* (§9)
- Quantiles: $x_p = \sigma z_p + \mu$

Examples: using the representation

Exponential with mean 2

$X = 2Z$ with $Z \sim \mathrm{Exp}(1)$.

- $P(X > 3) = P(Z > 1.5) = e^{-1.5} \approx 0.223$.
- Median: $2 \cdot \ln 2 \approx 1.386$.
- $\mathbb{E}X = 2 \cdot 1$, $\mathrm{Var}\, X = 4 \cdot 1$.



Cauchy(μ, σ)

$X = \sigma Z + \mu$, Z standard Cauchy.

- The median is μ (not a mean: none exists!).
- $P(|X - \mu| < \sigma) = P(|Z| < 1) = \frac{2}{\pi} \arctan 1 = \frac{1}{2}$.
- So $\mu \pm \sigma$ are the *quartiles*: σ is the interquartile half-range.

Takeaway

For location–scale families the parameters mean "center" and "spread" even when $\mathbb{E}$ and Var do not exist.

Activity 2 — Location–scale families

Problems

(a) Show that $g(x) = \frac{1}{\sigma} f\left(\frac{x-\mu}{\sigma}\right)$ with $f(z) = \frac{1}{2} e^{-|z|}$ is the double exponential(μ, σ). If $Z \sim f$, use $\mathbb{E}Z = 0$, $\operatorname{Var} Z = 2$ to find $\mathbb{E}X$ and $\operatorname{Var} X$ for $X = \sigma Z + \mu$.

(b) $X \sim \mathrm{Uniform}(5, 9)$. Write $X = \sigma Z + \mu$ with $Z \sim \mathrm{Uniform}(0, 1)$. Find μ, σ, then $\mathbb{E}X$, $\operatorname{Var} X$.

(c) $X \sim \mathrm{Cauchy}(3, 2)$. Find $P(1 < X < 5)$ and $P(X > 3)$.

(d) True or false: "Every location–scale family has a mean." Give a counterexample.

Activity 2 — Solutions

(a) $g(x) = \frac{1}{2\sigma} e^{-|x-\mu|/\sigma}$. $\mathbb{E}X = \sigma \cdot 0 + \mu = \mu$, $\operatorname{Var} X = \sigma^2 \cdot 2 = 2\sigma^2$.

(b) $\mu = 5$, $\sigma = 4$ ($X = 4Z + 5$). $\mathbb{E}X = 4 \cdot \frac{1}{2} + 5 = 7$, $\operatorname{Var} X = 16 \cdot \frac{1}{12} = \frac{4}{3}$.

(c) $X = 2Z + 3$: $P(1 < X < 5) = P(-1 < Z < 1) = \frac{1}{2}$; $P(X > 3) = P(Z > 0) = \frac{1}{2}$.

(d) False. Cauchy(μ, σ) is a location–scale family with no mean.

§3.6 Motivation: when exact probabilities are out of reach

Situations

- You only know $\mathbb{E}X$ and $\operatorname{Var} X$, not the distribution.
- The cdf has no closed form (e.g. the normal) but you need a quick, safe bound.
- A moment such as $\mathbb{E}X^6$ would need a nasty integral or sum.

Plan

Inequalities: Chebyshev-type bound, a normal tail bound.
Identities: Stein's lemma (normal) and its Poisson analogue: recursions for moments.

Quick quiz

A random variable has $\mathbb{E}X = 50$ and $\operatorname{Var} X = 25$. At least what fraction of its probability lies in $(40, 60)$?

(Chebyshev: $P(|X - 50| \geq 10) \leq \frac{25}{100}$, so at least 75%.)

Chebyshev's inequality (general form)

Theorem (CB 3.6.1)

Let X be a random variable and $g(x) \geq 0$ a nonnegative function. Then for every $r > 0$

$$P(g(X) \geq r) \leq \frac{\mathbb{E}\, g(X)}{r}.$$

Proof (continuous)

$$\begin{aligned}\mathbb{E}g(X) &\geq \int_{\{g \geq r\}} g(x) f(x) dx \\ &\geq r \int_{\{g \geq r\}} f(x) dx = rP(g(X) \geq r).\end{aligned}$$

Special cases

- $g(x) = x$ $(X \geq 0)$: **Markov**, $P(X \geq r) \leq \mathbb{E}X/r$.
- $g(x) = \frac{(x-\mu)^2}{\sigma^2}$, $r = t^2$: **Chebyshev**, $P(|X - \mu| \geq t\sigma) \leq \frac{1}{t^2}$.
- $g(x) = e^{sx}$: **Chernoff**, $P(X \geq a) \leq e^{-sa} M_X(s)$.

Example: the choice of g matters

$X \sim \text{Exponential}(1)$: bound $P(X \geq 4)$

Choice of g	**Bound**	**Value**
$g(x) = x$ (Markov)	$\mathbb{E}X/4$	0.250
$g(x) = x^2$	$\mathbb{E}X^2/16 = 2/16$	0.125
$g(x) = (x-1)^2$, $\{\lvert X-1\rvert \geq 3\} \supseteq \{X \geq 4\}$	$\text{Var}\, X/9$	0.111
$g(x) = e^{sx}$, best $s = \frac{3}{4}$ (Chernoff)	$4e^{-3}$	0.199
True value	e^{-4}	**0.0183**

Lessons

All bounds are valid but crude: Chebyshev-type inequalities use almost no information about the distribution. They are most useful for *theory* (Ch. 5), where a crude bound is enough, not for numerical probabilities.

A sharper bound for the normal

Normal tail inequality

For $Z \sim N(0,1)$ and $t > 0$:

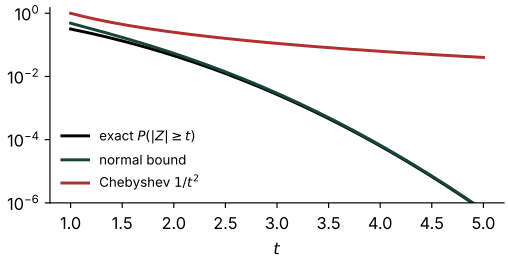
$$P(|Z| \geq t) \leq \sqrt{\frac{2}{\pi}} \, \frac{e^{-t^2/2}}{t}.$$

Proof

For $z \geq t$: $1 \leq z/t$. So

$$P(Z \geq t) \leq \int_t^\infty \frac{z}{t} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = \frac{e^{-t^2/2}}{t\sqrt{2\pi}}.$$

Double for $|Z|$ by symmetry.



t	exact	normal bd	Cheb.
1	0.317	0.484	1
2	0.0455	0.0540	0.25
3	0.0027	0.0030	0.111

Stein's lemma (a normal identity)

Lemma (Stein; CB 3.6.5)

Let $X \sim N(\theta, \sigma^2)$ and g a differentiable function with $\mathbb{E}|g'(X)| < \infty$. Then

$$\mathbb{E}\big[g(X)(X-\theta)\big] = \sigma^2\, \mathbb{E}\, g'(X).$$

Proof

The normal pdf satisfies $f'(x) = -\frac{x-\theta}{\sigma^2} f(x)$, i.e. $(x-\theta)f(x) = -\sigma^2 f'(x)$. Integrate by parts:

$$\mathbb{E}[g(X)(X-\theta)] = -\sigma^2 \int g(x) f'(x)\,dx = -\sigma^2 \big[gf\big]_{-\infty}^{\infty} + \sigma^2 \int g'(x) f(x)\,dx = \sigma^2 \mathbb{E} g'(X),$$

because $g(x)f(x) \to 0$ at $\pm\infty$ under the moment condition.

In words

Covariance of X with $g(X)$ equals σ^2 times the average slope of g. Differentiation replaces integration.

Stein's lemma: examples

Higher moments of $N(\theta, \sigma^2)$

Take $g(x) = (x-\theta)^n$, $g'(x) = n(x-\theta)^{n-1}$:

$$\mathbb{E}(X-\theta)^{n+1} = \sigma^2\, n\, \mathbb{E}(X-\theta)^{n-1}.$$

- $n=1$: $\mathbb{E}(X-\theta)^2 = \sigma^2$.
- $n=3$: $\mathbb{E}(X-\theta)^4 = 3\sigma^2 \cdot \sigma^2 = 3\sigma^4$.
- $n=5$: $\mathbb{E}(X-\theta)^6 = 5\sigma^2 \cdot 3\sigma^4 = 15\sigma^6$.

The normal mgf, again

$g(x) = e^{sx}$, $g' = se^{sx}$: $\mathbb{E}[(X-\theta)e^{sX}] = \sigma^2 s\, M(s)$, i.e.

$$M'(s) = (\theta + \sigma^2 s) M(s), \quad M(0) = 1.$$

Solving the ODE: $M(s) = e^{\theta s + \sigma^2 s^2/2}$.

Remark

Stein's lemma is also the starting point of Stein's method for normal approximation, and of shrinkage estimation (James–Stein).

A Poisson identity

Lemma (Poisson identity; CB 3.6.8)

Let $X \sim \text{Poisson}(\lambda)$ and g any function with $\mathbb{E}|g(X)| < \infty$. Then

$$\mathbb{E}\big[X\, g(X)\big] = \lambda\, \mathbb{E}\, g(X+1).$$

Proof

$$\mathbb{E}[Xg(X)] = \sum_{x=1}^{\infty} x\, g(x) \frac{e^{-\lambda}\lambda^x}{x!} = \sum_{x=1}^{\infty} g(x) \frac{e^{-\lambda}\lambda^x}{(x-1)!}$$
$$= \lambda \sum_{y=0}^{\infty} g(y+1) \frac{e^{-\lambda}\lambda^y}{y!} = \lambda \mathbb{E} g(X+1).$$

Moments of Poisson(λ)

- $g(x) = 1$: $\mathbb{E}X = \lambda$.
- $g(x) = x$: $\mathbb{E}X^2 = \lambda\mathbb{E}(X+1) = \lambda^2 + \lambda$, so $\text{Var}\, X = \lambda$.
- $g(x) = x^2$: $\mathbb{E}X^3 = \lambda\mathbb{E}(X+1)^2 = \lambda(\lambda^2 + \lambda + 2\lambda + 1) = \lambda^3 + 3\lambda^2 + \lambda$.

Summary of §3.4–3.6

Topic	Key statement	Use
Exponential family	$f = h(x)c(\theta)\exp\{\sum w_i(\theta)t_i(x)\}$; $\mathbb{E}t(X) = -\frac{d}{d\eta}\log c^*(\eta)$	moments without integrals
Location–scale	$X = \sigma Z + \mu$, $f_X = \frac{1}{\sigma} f(\frac{x-\mu}{\sigma})$	reduce to standard Z; pivots
Chebyshev	$P(g(X) \geq r) \leq \mathbb{E}g(X)/r$	crude bounds from moments
Normal tail bound	$P(\lvert Z\rvert \geq t) \leq \sqrt{2/\pi}\, e^{-t^2/2}/t$	sharper than Chebyshev
Stein's lemma	$\mathbb{E}[g(X)(X-\theta)] = \sigma^2\mathbb{E}g'(X)$	normal moments
Poisson identity	$\mathbb{E}[Xg(X)] = \lambda\mathbb{E}g(X+1)$	Poisson moments

Activity 3 — Inequalities and identities

Problems

1. $X \sim N(0,1)$. Use Stein's lemma to find $\mathbb{E}X^4$ and $\mathbb{E}X^6$.
2. $X \sim \text{Poisson}(2)$. Use the Poisson identity to find $\mathbb{E}X^3$.
3. $X \sim \text{Exponential}(1)$. Use Chebyshev's inequality with $g(x) = x^2$ to bound $P(X \geq 4)$. Compare with the exact value.
4. For $Z \sim N(0,1)$, compare Chebyshev and the normal tail inequality for $P(|Z| \geq 2)$.
5. Which family (if any) is an exponential family: Uniform$(0,\theta)$; Poisson(λ); Cauchy(θ)?

Activity 3 — Solutions

(a) $\mathbb{E}X^4 = 3 \cdot 1 \cdot \mathbb{E}X^2 = 3$ (Stein, $g = x^3$). $\mathbb{E}X^6 = 5\,\mathbb{E}X^4 = 15$.

(b) $\mathbb{E}X^3 = \lambda\mathbb{E}(X+1)^2 = 2(\mathbb{E}X^2 + 2\mathbb{E}X + 1) = 2(6+4+1) = 22$ (with $\mathbb{E}X^2 = \lambda^2 + \lambda = 6$).

(c) $P(X \geq 4) = P(X^2 \geq 16) \leq \mathbb{E}X^2/16 = 2/16 = 0.125$; exact $e^{-4} \approx 0.0183$.

(d) Chebyshev: $\leq \frac{1}{4}$. Normal bound: $\sqrt{2/\pi}\, e^{-2}/2 \approx 0.054$. Exact: 0.0455.

(e) Only Poisson(λ).

Next

Chapter 4: joint, marginal and conditional distributions; independence; covariance.